Hands-on

Threads

Python Training

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Threads

Step 1: Open Jupyter notebook and import these two packages:

```
from time import sleep
from threading import Thread
```

Step 2: Once you have imported the packages now write the below code for starting single thread:

```
def thread_function(name):
    print("Thread {0} started".format(name))
    sleep(2)
    print("Thread {0} ended".format(name))
print('Main - started')
t=Thread(target=thread_function, args=("New",))
print("Main - Before Starting Thread")
t.start()
print("Main- Waiting for thread to finish")
print("Main- Before Starting Thread")
print("Main- Done")
```

Step 3: Execute the code:

```
[6]: def thread_function(name):
         print("Thread {0} started".format(name))
         sleep(2)
         print("Thread {0} ended".format(name))
     print('Main - started')
     t=Thread(target=thread_function, args=("New",))
     print("Main - Before Starting Thread")
     t.start()
     print("Main- Waiting for thread to finish")
     print("Main- Before Starting Thread")
     print("Main- Done")
```

```
Main - started
Main - Before Starting Thread
Thread New startedMain- Waiting for thread to finish
Main- Before Starting Thread
Main- Done

Thread New ended
```

Starting Multiple thread:

Step 1:

Mention the below code for analyzing multiple threads:

```
def thread_function(name):
    print("Thread {0} started".format(name))
    sleep(2)
    print("Thread {0} ended".format(name))
threads=[]
for index in range(3):
    print("Main :: Creating and starting a thread {0}".format(index))
    t=Thread(target=thread_function, args=(index,))
    threads.append(t)
    t.start()
for index, thread in enumerate(threads):
    print("Main :: Before Joining thread {0}".format(index))
    thread.join()
    print("Main :: After joining thread {0}".format(index))
```

Check the output:

```
Main :: Creating and starting a thread 0
Thread 0 startedMain :: Creating and starting a thread 1

Thread 1 startedMain :: Creating and starting a thread 2

Thread 2 startedMain :: Before Joining thread 0

Thread 1 endedThread 0 ended

Main :: After joining thread 0
Main :: Before Joining thread 1
Main :: After joining thread 1
Main :: Before Joining thread 2
Thread 2 ended
Main :: After joining thread 2
```